

Supplementary Information

Cognate redocking as a quality-control step for cryo-EM kinase receptor structures: a study of ATR/CHK1 and CDK-activating kinase

Supplementary Section S1: Other CASF-2016 evaluations report higher top-pose success rates under a different protocol

The 58% figure cited in the main text (Methods, Results §3.3) is AutoDock Vina’s *self-docking* (redocking) success rate on the 285-complex PDBbind Core Set, as reported by Röhrig et al. [1]: the ligand is removed from its own co-crystallized receptor and re-docked from scratch, then scored against a ≤ 2.0 Å RMSD threshold to the native pose – the same protocol used throughout this study. This is a different, and considerably harder, test than CASF-2016’s own *docking power* metric, which asks whether the correct (near-native) pose is ranked at or near the top of a pre-generated, decoy-enriched pose ensemble rather than found by a full conformational search. Under that docking-power protocol, Su et al. [2] report that AutoDock Vina achieves a top-1 pose success rate of 90.2% (257 of 285 complexes) on CASF-2016. The two figures (58% self-docking vs. 90.2% docking power) are not directly comparable – they test different things – but the gap illustrates why the 58% self-docking baseline, not the 90.2% docking-power figure, is the appropriate comparator for this study’s redocking protocol, and why the 58% value should not be read as a general “Vina success rate.”

Supplementary Table S2: Residue-deletion list (flexible-sidechain receptor preparation)

Methods §2.1 and §2.8 report that Meeko’s receptor-preparation step automatically deleted approximately 400 residues from the 9L40 receptor whose side-chain density could not be matched against a template at the structure’s resolution, and that all deleted residues were >10 Å from the docking box. **This enumerated per-residue list is not currently available as a standalone data file in the project repository:** the deletions are performed programmatically by Meeko at receptor-preparation time and were not captured to a separate log or table when the pipeline was originally run. Reproducing this list exactly would require re-running `mk_prepare_receptor.py` on the 9L40 structure and diffing its output against the input receptor – the authors have not done this and are flagging the gap here rather than presenting a reconstructed or estimated list as original data. The prepared receptor and ligand PDBQT files used for the flexible-docking runs (from which this list could be regenerated) are available in the pipeline repository (Data availability, main text).

Supplementary Table S4: Complete developability-passed derivative ranking

The complete ranking (677 developability-passed derivatives, satisfying $\geq 3/4$ drug-likeness rules with no structural alert) is provided as `Table_S4_derivative_ranking.csv` alongside this document. The first 20 ranked entries are shown below for reference; compounds flagged

`selected_for_pilot_docking = yes` are the 35 derivatives that entered the 40-compound pilot library (Table 4, main text).

Note on the main-text count: the Methods section of the main text states 680 of 786 candidates satisfied $\geq 3/4$ rules; the current pipeline output (`developability_predictions.csv`) shows 677. This small discrepancy has not been resolved and is reported here rather than silently corrected in the main text.

Rank	Compound ID (if selected)	Mol. wt. (Da)	Rules passed	Selected
1	deriv_000	331.4	4/4	yes
2	deriv_001	332.4	4/4	yes
3	deriv_002	340.4	4/4	yes
4	deriv_003	341.4	4/4	yes
5	deriv_004	345.4	4/4	yes
6	deriv_005	345.4	4/4	yes
7	deriv_006	346.4	4/4	yes
8	deriv_007	348.4	4/4	yes
9	deriv_008	348.4	4/4	yes
10	deriv_009	349.4	4/4	yes
11	deriv_010	349.4	4/4	yes
12	deriv_011	349.4	4/4	yes
13	deriv_012	349.4	4/4	yes
14	deriv_013	350.4	4/4	yes
15	deriv_014	350.4	4/4	yes
16	deriv_015	350.4	4/4	yes
17	deriv_016	350.4	4/4	yes
18	deriv_017	351.4	4/4	yes
19	deriv_018	351.4	4/4	yes
20	deriv_019	354.4	4/4	yes

Supplementary Table S5: Flexible-sidechain redocking, individual replicate values

Individual-seed naive and symmetry-corrected RMSDs and Vina scores for the flexible-sidechain redocking run (five residues, 30 Å box, Meeko preparation; Methods §2.8, Results §3.4). The matched rigid control (same 30 Å box and Meeko-prepared receptor/ligand, no residues declared flexible) is reported in the main text and pipeline repository as a mean $\pm$ SD (4.50 ± 0.05 Å); its individual per-seed values were not retained as a separate structured output and are not reproduced here for the same reason given for Table S2.

Condition	Seed	Naive RMSD (Å)	Symm. RMSD (Å)	Vina score (kcal/mol)
Flexible (5 res., 30 Å box)	1000	4.5427	4.2882	-9.795
Flexible (5 res., 30 Å box)	1001	2.7159	2.2256	-9.547
Flexible (5 res., 30 Å box)	1002	2.774	2.4945	-9.233
Flexible (5 res., 30 Å box) mean $\pm$ SD			3.0028 $\pm$ 0.9155	

Supplementary Table S6: Individual replicate RMSDs underlying Tables 1–3

Symmetry-corrected RMSD for every seeded replicate (three seeds per structure) behind the summary values reported in Tables 1–3 of the main text: the three primary ATR/CHK1 structures (exhaustiveness 32), the 14 additional CAK survey structures, and the 20 CASF-2016 sanity-check complexes (all at exhaustiveness 16, per Methods).

Table	Structure	Seed	Symm. RMSD (Å)
Table 1 (primary, exhaustiveness 32)	9L40	1000	2.5334
Table 1 (primary, exhaustiveness 32)	9L40	1001	2.5183
Table 1 (primary, exhaustiveness 32)	9L40	1002	2.541
Table 1 (primary, exhaustiveness 32)	9L4B	1000	0.586
Table 1 (primary, exhaustiveness 32)	9L4B	1001	0.6217
Table 1 (primary, exhaustiveness 32)	9L4B	1002	0.6242
Table 1 (primary, exhaustiveness 32)	2YM8	1000	0.7503
Table 1 (primary, exhaustiveness 32)	2YM8	1001	0.7596
Table 1 (primary, exhaustiveness 32)	2YM8	1002	0.7737
Table 2 (17-structure survey)	8P6V	1000	6.1362
Table 2 (17-structure survey)	8P6V	1001	6.1415
Table 2 (17-structure survey)	8P6V	1002	6.1382
Table 2 (17-structure survey)	8P6W	1000	7.3217
Table 2 (17-structure survey)	8P6W	1001	7.3382
Table 2 (17-structure survey)	8P6W	1002	7.3151
Table 2 (17-structure survey)	8P6X	1000	6.5755
Table 2 (17-structure survey)	8P6X	1001	6.6673
Table 2 (17-structure survey)	8P6X	1002	6.5535
Table 2 (17-structure survey)	8P6Z	1000	2.1761
Table 2 (17-structure survey)	8P6Z	1001	6.1707
Table 2 (17-structure survey)	8P6Z	1002	6.1794
Table 2 (17-structure survey)	8P70	1000	6.2449
Table 2 (17-structure survey)	8P70	1001	6.2502
Table 2 (17-structure survey)	8P70	1002	6.2477
Table 2 (17-structure survey)	8P71	1000	7.0164
Table 2 (17-structure survey)	8P71	1001	7.0119
Table 2 (17-structure survey)	8P71	1002	7.0354
Table 2 (17-structure survey)	8P72	1000	2.6997
Table 2 (17-structure survey)	8P72	1001	2.6968
Table 2 (17-structure survey)	8P72	1002	2.7035
Table 2 (17-structure survey)	8P73	1000	5.8317
Table 2 (17-structure survey)	8P73	1001	5.842
Table 2 (17-structure survey)	8P73	1002	5.8357
Table 2 (17-structure survey)	8P74	1000	6.6669
Table 2 (17-structure survey)	8P74	1001	6.675
Table 2 (17-structure survey)	8P74	1002	6.6784
Table 2 (17-structure survey)	8P75	1000	1.0119
Table 2 (17-structure survey)	8P75	1001	1.0533
Table 2 (17-structure survey)	8P75	1002	1.0206
Table 2 (17-structure survey)	8P76	1000	6.6039
Table 2 (17-structure survey)	8P76	1001	6.5926
Table 2 (17-structure survey)	8P76	1002	6.5984
Table 2 (17-structure survey)	8P77	1000	3.0258
Table 2 (17-structure survey)	8P77	1001	6.0337
Table 2 (17-structure survey)	8P77	1002	6.0076
Table 2 (17-structure survey)	8P78	1000	2.9874
Table 2 (17-structure survey)	8P78	1001	2.9735
Table 2 (17-structure survey)	8P78	1002	2.9702
Table 2 (17-structure survey)	8P7L	1000	6.1469
Table 2 (17-structure survey)	8P7L	1001	6.1418
Table 2 (17-structure survey)	8P7L	1002	6.199
Table 3 (CASF-2016 sanity check)	1O3F	1000	1.4579
Table 3 (CASF-2016 sanity check)	1O3F	1001	1.4884
Table 3 (CASF-2016 sanity check)	1O3F	1002	1.4151
Table 3 (CASF-2016 sanity check)	1SYI	1000	11.8023
Table 3 (CASF-2016 sanity check)	1SYI	1001	11.8423
Table 3 (CASF-2016 sanity check)	1SYI	1002	11.8268
Table 3 (CASF-2016 sanity check)	2CET	1000	7.7526
Table 3 (CASF-2016 sanity check)	2CET	1001	7.7367
Table 3 (CASF-2016 sanity check)	2CET	1002	7.742
Table 3 (CASF-2016 sanity check)	2J7H	1000	9.3411
Table 3 (CASF-2016 sanity check)	2J7H	1001	9.3751

Table	Structure	Seed	Symm. RMSD (Å)
Table 3 (CASF-2016 sanity check)	2J7H	1002	9.3633
Table 3 (CASF-2016 sanity check)	2P4Y	1000	0.5712
Table 3 (CASF-2016 sanity check)	2P4Y	1001	0.4968
Table 3 (CASF-2016 sanity check)	2P4Y	1002	0.4805
Table 3 (CASF-2016 sanity check)	2W4X	1000	1.6605
Table 3 (CASF-2016 sanity check)	2W4X	1001	1.6689
Table 3 (CASF-2016 sanity check)	2W4X	1002	1.6799
Table 3 (CASF-2016 sanity check)	2XBV	1000	1.0321
Table 3 (CASF-2016 sanity check)	2XBV	1001	1.0494
Table 3 (CASF-2016 sanity check)	2XBV	1002	1.0399
Table 3 (CASF-2016 sanity check)	3ARQ	1000	3.1997
Table 3 (CASF-2016 sanity check)	3ARQ	1001	3.2081
Table 3 (CASF-2016 sanity check)	3ARQ	1002	3.1914
Table 3 (CASF-2016 sanity check)	3B65	1000	0.486
Table 3 (CASF-2016 sanity check)	3B65	1001	0.4871
Table 3 (CASF-2016 sanity check)	3B65	1002	0.4886
Table 3 (CASF-2016 sanity check)	3D6Q	1000	6.1551
Table 3 (CASF-2016 sanity check)	3D6Q	1001	6.1653
Table 3 (CASF-2016 sanity check)	3D6Q	1002	6.1546
Table 3 (CASF-2016 sanity check)	3DX1	1000	3.3101
Table 3 (CASF-2016 sanity check)	3DX1	1001	3.2738
Table 3 (CASF-2016 sanity check)	3DX1	1002	3.294
Table 3 (CASF-2016 sanity check)	3KWA	1000	3.3611
Table 3 (CASF-2016 sanity check)	3KWA	1001	3.3752
Table 3 (CASF-2016 sanity check)	3KWA	1002	3.5037
Table 3 (CASF-2016 sanity check)	3NQ9	1000	5.6598
Table 3 (CASF-2016 sanity check)	3NQ9	1001	9.8728
Table 3 (CASF-2016 sanity check)	3NQ9	1002	6.0619
Table 3 (CASF-2016 sanity check)	3PWW	1000	1.6869
Table 3 (CASF-2016 sanity check)	3PWW	1001	1.7456
Table 3 (CASF-2016 sanity check)	3PWW	1002	1.7894
Table 3 (CASF-2016 sanity check)	3WTJ	1000	6.3503
Table 3 (CASF-2016 sanity check)	3WTJ	1001	6.3478
Table 3 (CASF-2016 sanity check)	3WTJ	1002	6.3491
Table 3 (CASF-2016 sanity check)	4IH7	1000	4.5736
Table 3 (CASF-2016 sanity check)	4IH7	1001	4.572
Table 3 (CASF-2016 sanity check)	4IH7	1002	4.5719
Table 3 (CASF-2016 sanity check)	4K18	1000	4.5222
Table 3 (CASF-2016 sanity check)	4K18	1001	4.4707
Table 3 (CASF-2016 sanity check)	4K18	1002	4.5145
Table 3 (CASF-2016 sanity check)	4QAC	1000	0.5252
Table 3 (CASF-2016 sanity check)	4QAC	1001	0.5617
Table 3 (CASF-2016 sanity check)	4QAC	1002	0.5207
Table 3 (CASF-2016 sanity check)	4W9C	1000	9.5718
Table 3 (CASF-2016 sanity check)	4W9C	1001	9.6125
Table 3 (CASF-2016 sanity check)	4W9C	1002	9.5754
Table 3 (CASF-2016 sanity check)	4W9I	1000	1.2615
Table 3 (CASF-2016 sanity check)	4W9I	1001	1.4075
Table 3 (CASF-2016 sanity check)	4W9I	1002	1.4124

References

- [1] Ute F. Röhrig, Maïte Goullieux, Matthieu Bugnon, and Vincent Zoete. Attracting cavities 2.0: Improving the flexibility and robustness for small-molecule docking. *Journal of Chemical Information and Modeling*, 63(12):3925–3940, 2023. doi: 10.1021/acs.jcim.3c00054.
- [2] Minyi Su, Qifan Yang, Yu Du, Guoqin Feng, Zhihai Liu, Yan Li, and Renxiao Wang. Comparative assessment of scoring functions: The CASF-2016 update. *Journal of Chemical Information and Modeling*, 59(2):895–913, 2019. doi: 10.1021/acs.jcim.8b00545.